

Thirumalai Venkadesan V
AI Engineer

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Agile and Result-driven Data Science & Machine Learning Evangelist with career advancements of over 4+ years in implementing advanced statistical & predictive modeling techniques and machine learning to validate findings and build, maintain, and improve models using an iterative approach.

PROFESSIONAL PROFILE

- Good experience using Image processing Machine Learning, Deep Learning, Computer Vision, NLP, LLM, Generative AI models.
- Have used Python, NLP, LLM, Open CV, Rand Machine Learning for the product development.
- Strong in Statistics, EDA, Feature Engineering, Feature Scaling, ETL, Terraform.
- Experience in life-cycle agile software development.

TECHNICAL EXPERTISE

Expertise in Strategic: Predicting Modelling Forecasting Models Statistical Modelling Advanced Analytic Image Processing Pattern Recognition CNN VGG ANN RNN LSTM Unet Optimization YOLO ML Ops (CI/CD) MLflow Open-AI DALL E LLM RAG Word Embedding Data Visualization Web Scraping GIT Chat bot API OCR Stream-lit
Languages : Python, R, MatLab, MySQL,Big Data(Hadoop(cloudera),Apache Spark)
Frameworks/ Library : Keras, Scikit-learn, TensorFlow, PyTorch, NLTK, spaCy, Transformers, YOLO, LangChain, OCR, Mediapipe,CV2,skimage
API Framework: Flask, Django, Fast API
Databases : My SQL,MongoDB
Cloud: AWS, Azure
Tools & Version Control: Git Hub, Google Doc AI, Amazon Textract, Docker, Kubernetes, Terraform
Data Visualization: Seaborn, Matplotlib, Plotly, Power-Bi, Tablue, Looker Studio

EDUCATIONAL QUALIFICATION

- B.E–ECE, from Arjun college of technology- Anna University. Coimbatore.
- Diploma ECE, from Sri Sowdambika polytechnic College, State Board, Aruppukottai.

Company	Technology	Designation	Duration
Colan Info Tech Pvt Ltd	Python, ML, DL, NLP, Open CV, Terraform, LLM,Genai, AWS	Sr.Data Scientist	Jul 2023 to Dec 2024
Orane Healthcare Pvt Ltd	Python, ML , NLP, DL, AWS	Data Scientist	May 2022 to April 2023
Mirnah Technology	Python, OpenCV	OpenCV Developer	June 2021 to Feb 2022
Pantech Solutions	ImageProcessing,Signalprocessing, Deep Learning,OpenCV	Project Engineer	Nov 2019 to Nov 2020

Project : Text Multi-class Classification using BERT**Client: LSG(Local Self Government Department)**

- BERT-Based Multi-Class Text Classification: Designed and fine-tuned a BERT model to classify invoice descriptions into 16+ GNCCode classes, including tasks like predicting Invoice ID and Fee Rate with high accuracy.
- NLP and Model Optimization: Led efforts in data preprocessing, tokenization, BERT embedding, model training, and evaluation, ensuring optimal performance through advanced NLP techniques.
- Cloud Integration and Deployment: Streamlined data storage for cloud providers, ensuring secure and efficient data handling. Deployed the solution using Flask, Docker, and AWS for scalable and reliable operation.
- End-to-End Development: Managed the backend with Python and TensorFlow, integrated a front-end using HTML, and utilized GitHub for version control and collaboration
- Deployed the project in the AWS cloud with CI/CD.

Project : Attendance Monitoring System with Face Recognition**Client : Colan Infotech**

- Designed and implemented facial feature denoising and enhancement techniques, boosting accuracy for attendance monitoring using AlexNet and Haar Cascade Classifier.
- Built and trained an AlexNet-based facial recognition system for real-time attendance tracking, ensuring robust performance and scalability.
- Integrated the system with Flask API on a Windows machine and deployed it on Docker and Heroku, enabling seamless functionality and scalability.
- Leveraged cutting-edge libraries such as OpenCV, Keras, TensorFlow, NumPy, and dlib to optimize the model's efficiency and performance.

Project : Intelligent Drilling Data Management and Predictive Analysis for Gold Exploration For Geologist Gold Mining and AI Summarization for Gold Mining**Client : Discovery Alert**

- Led the development of an AI-based system to classify documents as 'drilling' or 'non-drilling' and extract key drilling-related information such as Project Name, Country, and Best Drill Result using GPT-4.0 and LLM techniques.
- Designed a Geological Analysis Report (GAR) tool and a report drafting assistant, streamlining report generation and enhancing educational resources for geological analysis.
- Implemented Hold ID-based cross-section image extraction and prospectivity rating calculations using GPT-4.0 Vision Preview, improving accuracy and project documentation.
- Utilized GPT-4.0 Vision Preview to automate content creation for social media, significantly boosting engagement and visibility.
- Deployed solutions using Python, SQLAlchemy, and AzureOpenAI, integrated with front-end technologies like HTML, CSS, and JavaScript, and deployed on Azure Cloud Service for scalable operations.
- Streamlined data classification and extraction processes, automated reporting, enhanced educational resources, and improved communication with AI-generated text and image summaries.

Project Title: Intelligent School Management System with NLP-Powered ChatBot Integration Technologies**Client : Dubai**

- Designed a chatbot system with modules for Student, Attendance, Management, Parent, and Cafeteria, streamlining communication and database monitoring within the school community.
- Enhanced the chatbot experience in the second phase by incorporating 2D image avatars synchronized with audio - Speech capabilities, providing an engaging and interactive communication platform.
- Adopted a modular architecture to efficiently integrate multiple functionalities, ensuring seamless operation and scalability of the chatbot system.

- Leveraged libraries like TensorFlow, Keras, Scikit-learn, and GenAI for implementing advanced audio-to-speech synchronization and AI-driven responses.
- Deployed the system using Flask API, ensuring accessibility and seamless integration, while continuous feedback and refinement improved usability and effectiveness.

Project Title: AI-Powered Dental CRM with LLM-Based Virtual Agent

Start-up Product Based Project link: growdentalcrm.com

- Developed an AI-powered **Dental CRM** for booking, canceling, and rescheduling appointments with seamless doctor and patient interactions.
- Designed a **Doctor Agent** to enable doctors to access existing patient records and manage appointments efficiently.
- Built a **Patient Agent** that allows patients to view doctor availability and book appointments using **LLM (Mistral 7B) and Lang-chain** for natural language-based interactions.
- Integrated **PostgreSQL** for securely storing and updating patient data upon appointment creation, ensuring robust database management.
- Implemented **FuzzyWuzzy** for close-match name identification, correcting potential spelling mistakes in patient entries.
- Deployed the system using Flask API, enhancing accessibility, security, and real-time appointment tracking.

Project Title: JeeBuddy AI-Powered Educational Assistant

Start-up Product based Project link: jeebuddy.in

- Developed **JeeBuddy.in**, an AI-driven educational platform leveraging multiple models like **03 Mini, Mistral 7B, and Groq** for efficient Q&A-based learning.
- Built a **PDF and book-based Q&A system** that retrieves answers using **LLM prompting** without relying on **VectorDB**, ensuring direct and accurate responses.
- Enabled **student PDF uploads**, allowing users to select specific text and ask questions based on highlighted sections.
- Implemented **math formula explanation** using **symbolic reasoning** and NLP-based step-by-step breakdowns.
- Integrated **image analysis** with **base64 encoding**, applying **DeepSeek-V3 in parallel** for deeper insights while leveraging **GPT-4.0 Vision for enhanced image understanding**.
- Developed a **video generation system** using **Manim** to create **Mathematics and Physics animations** based on AI-generated scripts.
- Deployed the system on **AWS**, optimizing performance with **NGINX for faster response times** and scalable infrastructure.

Declaration

I hereby declare that the information endowed above is genuine to the best of my knowledge and belief.

(THIRUMALAIVENKADESAN)